Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Which of the following reactions will give bleaching powder-

(A) $\mathrm{CaCl_2 + H_2O}$

(B) $\mathrm{CaO + HCl}$

(C) $\mathrm{Ca(OH)_2 + Cl_2}$

(D) $\mathrm{ClO_2 + Ca(OH)_2}$

Q.2

An aq solution of a gas X gives the following reactions:
(i) It decolourises an acidified $\mathrm{K_2Cr_2O_7}$ solution
(ii) On boiling with $\mathrm{H_2O_2}$, cooling it and then adding an aq solution of $\mathrm{BaCl_2}$, a precipitate insoluble in dilute HCl is obtained.
(iii) On passing $\mathrm{H_2\ S}$ in the solution, white turbidity is obtained. Then X is:

(A) $\mathrm{SO_2}$

(B) $\mathrm{SO_3}$

(C) $\mathrm{CO_2}$

(D) CO

Q.3

The ease of liquefaction of noble gases increases in the order -

(A) $\mathrm{He < Ne < Ar < Kr < Xe}$

(B) $\mathrm{Xe < Kr < Ne < Ar < He}$

(C) $\mathrm{Kr < Xe < He < Ne < Ar}$

(D) $\mathrm{Ar < Kr < Xe < Ne < He}$

Q.4

When copper is reacted with dil. $\mathrm{HNO_3}$ a colourless gas is liberated. Which of the following is true for the gas?

(A) Has 11 valence e^-, where extra electron occupies an antibonding $\pi * 2p$ orbital.

(B) An acid anhydride.

(C) Paramagnetic in solid state.

(D) Has no resonance.

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Q.5

Cl_2Br_2 and I_2 react with $Na_2S_2O_3$. Pick out the correct statement from the following.

(A) All the halogens react in a similar way

(B) Cl_2 and Br_2 react in a similar way but I_2 react differently

(C) Br_2 and I_2 react in a similar way

(D) Cl_2 reacts differently

Q.6

Incorrect statement among the following is

(A) PCl_3 is prepared by action of thionyl chloride on white phosphorous.

(B) Partial hydrolysis of XeF_6 is a redox reaction.

(C) Catalyst used in Deacon's process is $CuCl_2$

(D) $FeCl_2$ is more ionic than $FeCl_3$.

Q.7

What is the hybrid state and oxiation state of sulphur in Caro's acid?

(A) $sp^2, +10$

(B) $sp^3, +10$

(C) $sp^2, +6$

(D) $sp^3, +6$

Q.8

Identify the correct sequence of increasing number of π-bonds in structures of the following molecules -
(i) $H_2S_2O_6$
(ii) H_2SO_3
(iii) $H_2S_2O_5$

(A) I, II, III

(B) II, III, I

(C) II, I, III

(D) I, III, II

Q.9

Identify the incorrect statement

(A) SF_6 is known but not SCl_6

(B) There are two ' $d\pi - p\pi$ ' bonds in SO_2

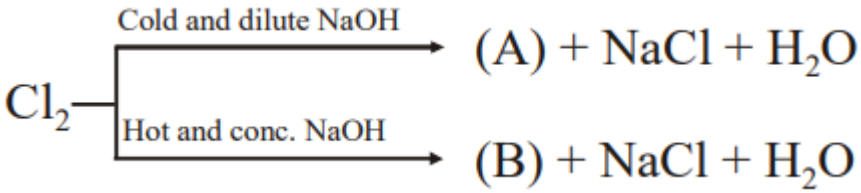
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(C) In $\gamma - SO_3$ there are six '$S - O'$ and '$S = O$' bonds

(D) SO_2 is more powerful reducing agent in alkaline medium than in acidic medium.

Q.10

Compounds (A) and (B) are -



(A) $NaClO_3, NaClO$

(B) $NaOCl_2, NaOCl$

(C) $NaClO_4, NaClO_3$

(D) $NaOCl, NaClO_3$

Q.11

When chlorine water is added to an aqueous solution of sodium iodide in the presence of chloroform, a violet colouration is obtained. On adding more of chlorine water and vigorous shaking, the violet colour disappears. This shows the conversion of.....into.....

(A) I_2, HIO_3

(B) I_2, HI

(C) HI, HIO_3

(D) I_2, HIO

Q.12

A greenish yellow gas reacts with an alkali hydroxide to form a halate which can be used in fireworks and safety matches. The gas and the halate are -

(A) $Br_2, KBrO_3$

(B) $Cl_2, KClO_3$

(C) $I_2, NaIO_3$

(D) I_2, KIO_3

Q.13

Which blue liquid is obtained on reacting equimolar amounts of two gases at $-30^\circ C$

(A) N_2O

(B) N_2O_3

(C) N_2O_4

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(D) N_2O_5

Q.14

Which one of the following statements regarding helium is/are correct

(A) It is used to produce and sustain powerful superconducting magnets

(B) It is used as a cryogenic agent for carrying out experiments at low temperatures

(C) It is used to fill gas balloons instead of hydrogen because it is lighter and non-inflammable

(D) It is used in gas-cooled nuclear reactors

Q.15

Identify the correct sequencing of hydrides based on the parameters specified in bracket:

(A) $NH_3 > PH_3 > AsH_3 > SbH_3$ (bond angle)

(B) $SbH_3 > NH_3 > AsH_3 > PH_3$ (boiling point)

(C) $NH_3 > PH_3 > AsH_3 > SbH_3$ (dipole moment)

(D) $NH_3 > PH_3 > AsH_3 > SbH_3$ (Lewis basic nature)

Q.16

Pick out correct statements -

(A) Orthohophorus acid acn be obtained by reacting P_4O_6 with H_2O

(B) Orthophosphoric acid can be obtained by reacting P_4O_{10} with H_2O

(C) Pyrophosphoric acid can be obtained by heating orthophosphourus acid

(D) Metaphosphoric acid is obtained by the dehydration of orthophosphoric acid at $316^\circ C$.

Q.17

Passage

Ozone is triatomic allotropic form of oxygen. It is an unstable, dark blue diamagnetic gas. O_3 is thermodynamically unstable and decomposes to O_2 $(O_3 \longrightarrow O_2 + O)$.

Due to ease with which it liberates atoms of nascent oxygen, it acts as a powerful oxidising agent, second only to F_2 in oxidising power.

A binary compound of oxygen with another element is called oxide. The oxides vary widely in their nature and properties.

Question

$2KOH + O_3 \longrightarrow [X] + O_2 + H_2O$

[X] is -

(A) K_2O_3

(B) KO_3

(C) K_2O

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(D) K_2O_2

Q.18

Passage

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Question

The correct order of acidic strength is -

(A) $K_2O < CaO < SiO_2 < CuO < CO_2 < N_2O_5$

(B) $K_2O < CaO < CuO < SiO_2 < CO_2 < N_2O_5$

(C) $CaO < K_2O < SiO_2 < CuO < CO_2 < N_2O_5$

(D) $K_2O < SiO_2 < CaO < CuO < CO_2 < N_2O_5$

Q.19

Passage

Ozone is triatomic allotropic form of oxygen. It is an unstable, dark blue diamagnetic gas. O_3 is thermodynamically unstable and decomposes to O_2 $(O_3 \longrightarrow O_2 + O)$.

Due to ease with which it liberates atoms of nascent oxygen, it acts as a powerful oxidising agent, second only to F_2 in oxidising power.

A binary compound of oxygen with another element is called oxide. The oxides vary widely in their nature and properties.

Question

Which of the following statements is incorrect -

(A) O_3 is used as disinfectant

(B) NO_2 is oxidised to N_2O_5 by O_3

(C) O_3- is paramagnetic in nature

(D) Dry iodide reacts with ozone and form I_2O_5

Q.20

Passage

About 95% of the earth's crust is composed of silicate, aluminosilicate clays, or silica. The majority of silicate minerals are very insoluble, because they have an infinite ionic structure and because of the great strength of the $Si - O$ bond. The basic structural units of silicate is $(SiO_4)^{4-}$ tetrahedra it may polymerize into larger units by sharing atoms.

Question

Proxenes and amphiboles cleave readily parallel to the chains forming fibers because -

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(A) the Si - O bonds in the chains are strong and directional

(B) The adjacent chains are held together by the metal ions present

(C) proxene is single chain silicate while amphibole is a double chain silicate

(D) (A) and (B) both

Answers & Solutions

Q.1 Answer:

$\mathrm{Ca(OH)_2 + Cl_2}$

Solution:

$$\mathrm{Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O}$$

Q.2 Answer:

$\mathrm{SO_2}$

Solution:

$$\mathrm{SO_2 \underset{K_2Cr_2O_7/H^+}{\longrightarrow} \textbf{Green } Cr^{3+}}$$

$$\mathrm{SO_3^{2-} \xrightarrow{H_2O_2} SO_4^{2-} \xrightarrow{BaCl_2} \textbf{White ppt } (BaSO_4)}$$

$$\mathrm{SO_2 + H_2\,S \rightarrow \underset{\left(\begin{array}{c}\text{white}\\ \text{thubidity}\end{array}\right)}{S} + H_2O}$$

Q.3 Answer:

$\mathrm{He < Ne < Ar < Kr < Xe}$

Solution:

As size increases, Vander Waals forces also increases. So, less is the ease of liquefaction.

Q.4 Answer:

Has 11 valence e^-, where extra electron occupies an antibonding $\pi * 2p$ orbital.

Solution:

$\mathrm{3Cu + 8HNO_3 \longrightarrow 2NO + 3Cu\,(NO_3)_2 + 4H_2O}$

$\mathrm{NO} \Rightarrow$ **has 11 valence** $\mathrm{e^-}$ **with an unpaired.**

$\mathrm{e^-}$ **in** π * $\mathrm{2p}$ **orbital.**

$\Rightarrow$ **is diamagnetic in solid state due to dimerisation.**

$\Rightarrow$ **shows resonance.**

Q.5 Answer:

$\mathrm{Cl_2}$ and $\mathrm{Br_2}$ react in a similar way but $\mathrm{I_2}$ react differently

Solution:

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$$\left.\begin{array}{l} Na_2S_2O_3 + Cl_2 + H_2O \longrightarrow Na_2SO_4 + 2HCl + S \\ Na_2S_2O_3 + Br_2 + H_2O \longrightarrow Na_2SO_4 + 2HBr + S \end{array}\right] \text{Similar reaction}$$

$$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI] \text{Different reaction}$$

Q.6 Answer:

Partial hydrolysis of XeF_6 is a redox reaction.

Solution:

Hydrolysis of X_eF_6 gives

$$XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$$

$$XeFF_6 + 2H_2O \rightarrow XeO_2\,F_2 + 4HF$$

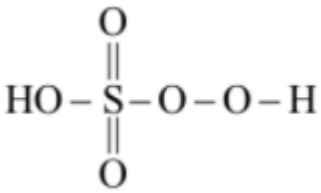
In above two reactions there is no change in oxidation number

Q.7 Answer:

$sp^3, +6$

Solution:

The structure of Caro's acid is

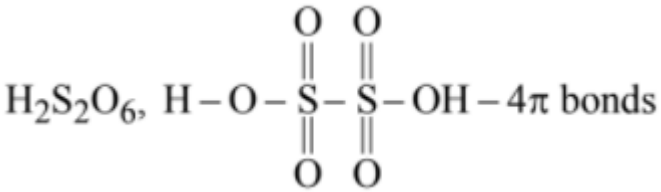


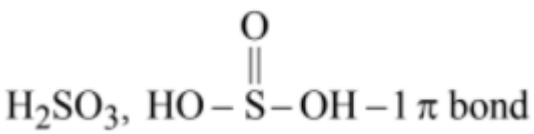
The hybrid state is sp^3 and its O.N. is $+6$.

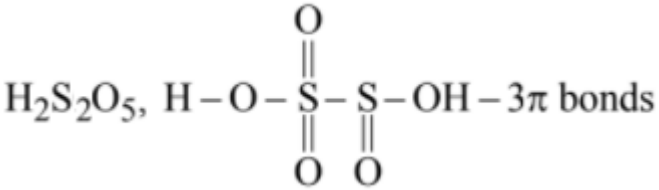
Q.8 Answer:

II, III, I

Solution:







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Q.9 Answer:

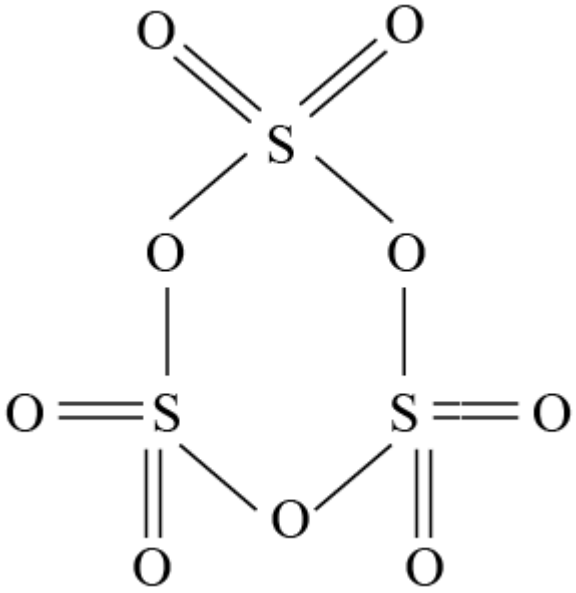
There are two ' $d\pi - p\pi$ ' bonds in SO_2

Solution:

1) SF_6 **is known but not** SCl_6 **because of small size of Fluorine**

2) SO_2 **has one** $d\pi - p\pi$ **bond**

3) $\gamma - SO_3$ **is**



There are 6 'S-O' & 'S=O' bond

Q.10 Answer:

$NaOCl, NaClO_3$

Solution:

$$Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$$

Cold and dil.

$$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$$

Hot and conc.

Q.11 Answer:

I_2, HIO_3

Solution:

$$Cl_2 + 2NaI \rightarrow 2NaCl + I_2$$

(Violet colouration in chloroform)

$$5Cl_2 + 6H_2O + I_2 \rightarrow 2HIO_3 + 10HCl$$

colourless

Q.12 Answer:

$Cl_2, KClO_3$

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Solution:

The halate used in fireworks and safety matches is $KClO_3$. Thus, the gas is Cl_2. $3Cl_2 + 6KOH \rightarrow KClO_3 + 5KCl + 3H_2O$

greenish

yellow gas

Q.13 Answer:

N_2O_3

Solution:

$$NO(g) + NO_2(g) \xrightarrow{-30^\circ C} \underset{\text{(Blue)}}{N_2O_3(\ell)}$$

Q.14 Answer:

It is used to produce and sustain powerful superconducting magnets

Solution:

Helium is twice as heavy as hydrogen and has a 92% lower lifting capacity. Helium has the lowest melting and boiling points of all the elements, making liquid helium an excellent coolant for many applications requiring extremely low temperatures, such as cryogenic research and superconducting magnets. In order to cool down super conduction magnets, helium is employed. This is necessary for them to function. It also serves as protection for welding firms. The same process is used to create silicon, germanium, zirconium, titanium, and zircon.

Q.15 Answer:

$NH_3 > PH_3 > AsH_3 > SbH_3$ (bond angle)

Solution:

Abcd

NH_3 PH_3 AsH_3 SbH_3

107° 93.7^0 91.8^0 91.3° **(bond angle)**

1.46D $0.55D$ $0.22D$ $0.12D$ **(dipole moment)**

SbH_3 NH_3 AsH_3 PH_3

$-18^\circ C - 33^\circ C - 62^\circ C - 88^\circ C$ **(boiling point)**

Q.16 Answer:

Orthohophorus acid acn be obtained by reacting P_4O_6 with H_2O

Solution:

(1) Considering phosphorous acid,

It is obtained by dissolving phosphorous trioxide in water

$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$

(2) Orthophosphoric acid,

It is formed when phosphorus pentoxide is boiled with hot water.

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$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$

(3) Pyrophosphoric acid,

It is prepared by heating orthophosphoric acid at $250^\circ C - 260^\circ C$

$2H_3PO_4 \longrightarrow H_4P_2O_7 + H_2O$

(4) Metaphoric acid,

It is prepared by heating orthophosphoric acid or pyrophosphoric acid or diammonium hydrogen orthophosphate.

$H_3PO_4 \overset{600CC}{\longrightarrow} HPO_3 + H_2O$

$H_4P_2O_7 \overset{600CC}{\longrightarrow} 2HPO_3 + H_2O$

∴ Option (4) is the incorrect statement.

Q.17 Answer:

KO_3

Solution:

(i) $2KOH + 5O_3 \longrightarrow 2KO_3$ **(Orange colour)** $+5O_2 + H_2O$

(ii) As difference in electronegativity between element and oxygen increases, acidic character decreases and basic character increases.

(iii)

(A) Correct statement

(B) $2NO_2 + O_3 \longrightarrow N_2O_5 + O_2$

(C) O_3^- **is paramagnetic**

(D) $2I_2 + 9\,[O_3] \longrightarrow I_4O_9$ **(yellow solid)** $+9O_2$

Q.18 Answer:

$K_2O < CaO < CuO < SiO_2 < CO_2 < N_2O_5$

Solution:

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Q.19 Answer:

Dry iodide reacts with ozone and form I_2O_5

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Q.20 Answer:

(A) and (B) both

Solution:

(A) and (B) statements are correct explanation of the question.

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